STAFF REVIEW · H446 082 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Merge two individually sorted runs.

2. Selected value used to partition an array.

3. No; subarrays still need sorting.

4. [7,1,4].

I Do — source worked steps

1. Method and result — Lomuto partition moves 2 and 1 left, pivot3 to index2 → [2,1,3,6,5].

We Do — guided CHECK

1. Method and result
Smaller 3,2 move left; pivot5 at index2 → [3,2,5,8,6].

You Do — source mark schemes

1. Moves2 then1 [2], partition [2,1,4,5,7] [3], pivot index2 [1].

2. Left indexes0–1 [2], right3–4 [2], sorted result [1,2,4,5,7] [2].

3. Unbalanced partitions n−1 and0 [2], O(n^2) time [2], up to O(n) recursion depth [2]; balanced partitions tend O(log n) depth.

Exit Ticket — source answer

1. False. Both sides may still need sorting.